JAMES MITCHELL

Machine Learning Engineer

Email: james.mitchell@gmail.com

Phone: +44 7911 456823

LinkedIn: linkedin.com/in/jamesmitchell-ml

GitHub: github.com/jamesmitchell

PROFESSIONAL SUMMARY

Senior Machine Learning Engineer with 7 years of experience building and deploying production ML systems across fintech and e-commerce sectors. I've led teams through complex model development cycles and have a track record of reducing inference latency by 40% through thoughtful system design. I'm particularly interested in solving real-world problems where ML can deliver measurable business impact, and I enjoy mentoring junior engineers through the messy parts of model development that textbooks don't cover.

WORK EXPERIENCE

Senior Machine Learning Engineer

DataFlow Systems Ltd

glasgow.dataflow.io

Glasgow, United Kingdom

March 2022 - Present

Designed and implemented a real-time recommendation engine serving 2 million daily active users, achieving 15% improvement in click-through rates through careful feature engineering and A/B testing methodology.

Led a team of 3 junior engineers in building an automated model monitoring pipeline that reduced incident response time from 4 hours to 15 minutes by catching data drift before it impacted predictions.

Refactored our legacy batch prediction system into a containerized microservices architecture using Docker and Kubernetes, cutting operational costs by 35% and enabling deployment frequency to increase from monthly to weekly.

Spearheaded adoption of MLflow for experiment tracking across the organization, which improved reproducibility and reduced time spent on duplicate work by roughly 20 hours per engineer annually.

Machine Learning Engineer

FinanceAI Solutions

financeai.co

Edinburgh, United Kingdom

July 2019 - February 2022

Built fraud detection models using gradient boosting that achieved 94% precision while maintaining 87% recall, directly preventing GBP 2.3 million in fraudulent transactions over 18 months.

Developed an internal Python library for time series feature engineering that standardized our approach across 5 different ML teams and reduced code duplication significantly.

Collaborated with data engineering and product teams to design and implement a feature store infrastructure that reduced model training time from 6 hours to 40 minutes through intelligent caching strategies.

Conducted code reviews and pair programming sessions with team members, identifying performance bottlenecks and suggesting optimizations that improved model inference speed by an average of 38%.

Machine Learning Engineer

ByteScale Analytics

bytescale.net

Glasgow, United Kingdom

June 2017 - June 2019

Developed and deployed clustering models to segment 50 million customer records, enabling the marketing team to run personalized campaigns that increased conversion rates by 22%.

Implemented automated data validation pipelines using Great Expectations that caught quality issues upstream, reducing the number of production incidents related to bad data by 78% within 6 months.

Participated in on-call rotations and maintained SLAs of 99.5% uptime for prediction APIs serving critical business dashboards used by the executive team.

Contributed to open source projects including scikit-learn and pandas, gaining experience working in large distributed codebases and understanding the perspectives of library maintainers.

EDUCATION

MSc Machine Learning

University of Edinburgh

Edinburgh, United Kingdom

2015 - 2017

BSc Computer Science

University of Glasgow

Glasgow, United Kingdom

2011 - 2015

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, Bash

Machine Learning: scikit-learn, XGBoost, TensorFlow, PyTorch, LightGBM, model evaluation and validation

Data Engineering: Apache Spark, SQL optimization, ETL pipeline design, feature engineering

MLOps and Deployment: Docker, Kubernetes, MLflow, Jenkins CI/CD, model monitoring and drift detection

Databases: PostgreSQL, Redis, Elasticsearch, MongoDB

Cloud Platforms: AWS (SageMaker, EC2, S3), Google Cloud (BigQuery, Vertex AI)

Visualization: Matplotlib, Seaborn, Plotly, Tableau

Tools and Practices: Git, Jupyter, Linux, statistical testing, A/B testing methodology

PUBLIC REPOSITORIES

github.com/jamesmitchell/time-series-anomaly-detector - Python package for detecting anomalies in time series data using isolation forests and LSTM autoencoders.

github.com/jamesmitchell/feature-store-utilities - Collection of utilities for managing feature engineering pipelines and caching strategies in production systems.

github.com/jamesmitchell/model-evaluation-framework - Framework for comprehensive model evaluation including fairness metrics, calibration analysis, and cross-validation strategies.

github.com/jamesmitchell/fraud-detection-benchmark - Reproducible benchmark comparing multiple fraud detection approaches on public datasets with detailed evaluation reports.